

StudyBuddy

APPLIED GENERATIVE AI · FINAL
PROJECT

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Problem

- Studying is passive, generic and does not deal with your weaknesses:



Re-reading isn't learning:
Underlining and re-reading feels productive but it is not.



Quizzes are standard across: Standard question banks are the same for everyone, rather than being specific to ones weakness



No memory across sessions: Most tools does not track your weak spots, hence give you the same material over and over again

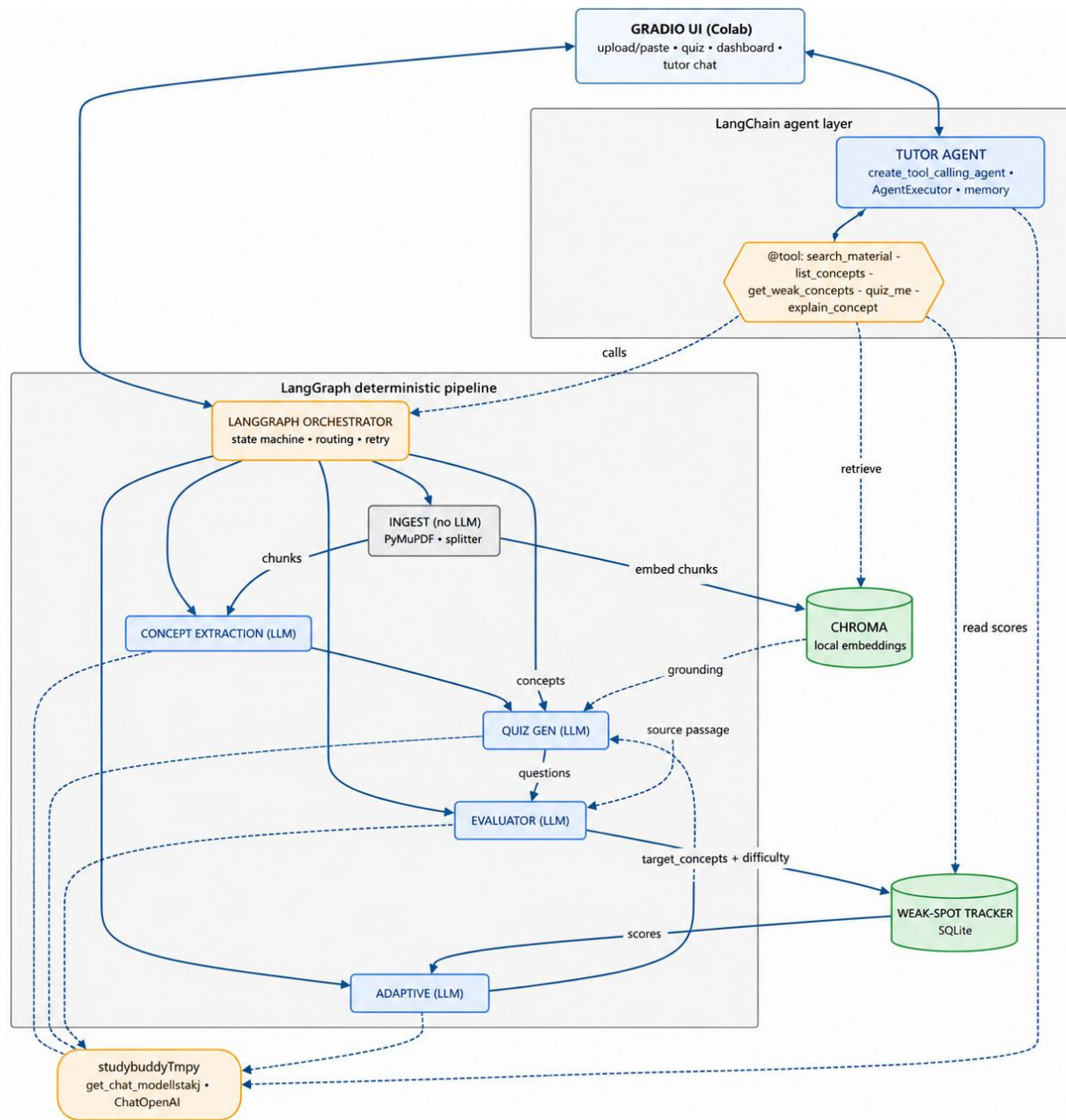


No tutor for own material: Online tools runs quizzes based on online sources not your lecture notes.



StudyBuddy

- Steps:
 1. Upload: PDF or notes are uploaded and split into chunks, embedded and mined for key concepts.
 2. Quiz: A grounded MCQ or True/False or short answer questions at 3 difficulty levels.
 3. Grade: Answers are checked against the source, with an explanation.
 4. Adapt: Weak concepts are reintroduced automatically and varies in difficulty along the progress



System architecture

- LangChain agent Layer:
 - The tool calling tutor accesses both stores and pipelines through “@tool” functions.
- LangGraph pipeline:
 - Ingest→Concept→Quiz→Evaluator→Adaptive, with routing and retry-on-incorrect form.
- Stores:
 - SQLite (the weak-spot tracker)
 - Chroma (local embeddings, RAG grounding)
- Model:
 - “get_chat_model(task)” in llm.py routes every LLM agent to its configured model.

Components

Component	Model	Function
Ingest	no LLM	Takes in PDFs or text files, splits them into topic-based sections using PyMuPDF and RecursiveCharacterTextSplitter, and stores them as embeddings in Chroma.
Concept Extraction	EXTRACTION_MODEL	Provides organized outputs by analyzing each chunk and extracting important phrases, definitions, and relationships using a specialized extraction model.
Quiz Generation	QUIZ_MODEL	Creates multiple-choice, true/false, or short-answer questions at three distinct difficulty levels by feeding the extracted concepts and background data into a quiz model.
Evaluator	EVAL_MODEL	Provides a conclusion and an explanation after comparing user responses to the original text using an evaluation model. This is done in code for multiple-choice questions and true/false; for short answers, it compares the answer to the source directly.
Weak Spot Tracker	no LLM	Uses SQLite to track user performance by concept, keeping an eye on tries, right answers, and streaks. There is no use of a linguistic model.
Adaptive	ADAPTIVE_MODEL	Uses the user's score table to generate a customized review plan. If the score is less than 60%, it suggests re-drilling; if it is more than 80%, it raises the difficulty; if not, it maintains the same level of difficulty.
Tutor (tool calling)	TUTOR_MODEL	Manages chat messages and has the ability to summon up to 10 different tools in order to offer grounded responses or trigger app operations when necessary.
Orchestrator	router	Uses LangGraph's StateGraph to route jobs through the pipeline; if any model produces output that is not correct, it automatically retries.

Models and routing

- One factory, many models



Concept Extraction:
GPT-4o



Quiz Generation:
Claude 3.5 Sonnet



Evaluator: Gemini
1.5 Flash



Adaptive: Claude 3
Haiku



Tutor: GPT-4o

Adaptability & Data



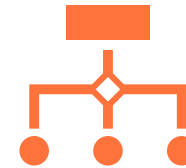
Grounded response:

Chunks are embedded into Chroma; your notes, not the public internet, provide the answers; the tutor and grader get the exact source passage and cite it.



Memory across sessions:

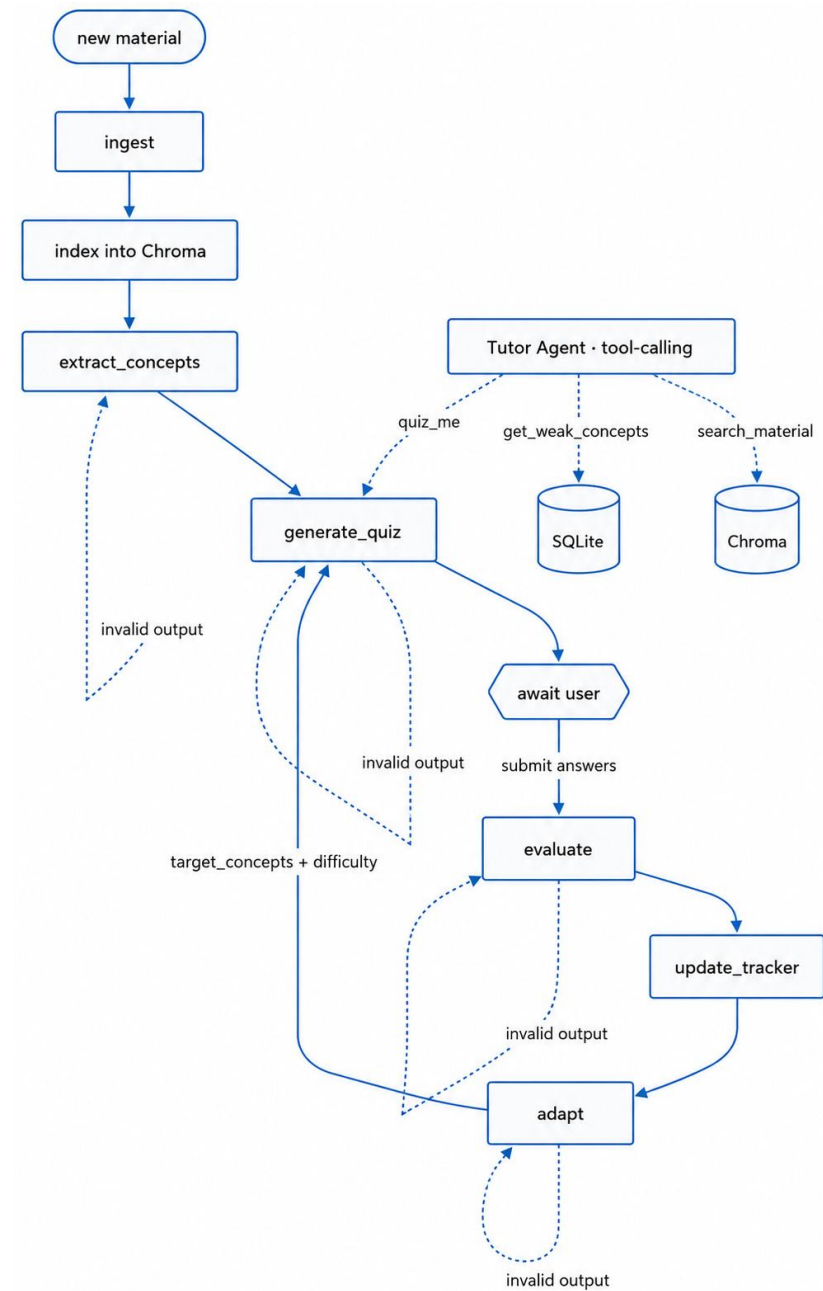
Per-concept tries, accurate counts, and streaks are stored in SQLite; the tutor uses RunnableWithMessageHistory to record conversation history for each session.



Adaptive loop:

The Adaptive agent analyzes the score table after grading; if it is less than 60%, re-drill; if it is more than 80%, increase the challenge. These targets are fed back into the quiz generation process by the LangGraph loop.

Adaptive Loop





User Interface

- A four card Gradio app
 - Upload:
 - Paste notes or drop a PDF. Concepts are ingested, embedded, and extracted. Handle several documents in a single session.
 - Quiz:
 - Provide feedback on each question in the current round. Control the number, question type, and complexity of the questions. Add more questions or regenerate them.
 - Dashboard:
 - Red, yellow, and green signs on a concept mastery map. has an accuracy trend, a "start weakness round" button, and Markdown export.
 - Tutor:
 - Chat about your content. Use cited RAG in your open-ended questions or give the instructor instructions like "quiz me on mitochondria." The agent selects the appropriate instrument.

Limitations

Limitations:

- StudyBuddy is not an authorized grader; rather, it is an educational tool. Model inaccuracies may be present in generated questions and grades.
- Because OCR is not covered, scanned or image-only PDFs without a text layer will not be extracted.
- The tutor requires a model that can make tool calls. It becomes just ordinary talk without one.
- Chroma is local and SQLite is single-writer, which is good for the demo but not the best for a large number of users.

For future:

- For scanned content, add OCR.
- Support richer document formats like PPTX and DOCX.
- For multi-user and multi-class scaling, switch to Postgres and a hosted vector database.
- Incorporate a more robust spaced-repetition schedule and confidence-weighted learning.
- To automatically score quizzes and grade quality, use the LangSmith evaluation.

A blue ballpoint pen with a silver-colored tip and clip is positioned diagonally across the frame. It rests on a document featuring a bar chart with several blue bars of varying heights. The background is a light blue grid. In the upper right corner, there is a small blue horizontal line.

Demo & Results

APPLICATION AND RESULTS ARE
IN THE ABOVE LINK

Thank you

